

Data Overview

The dataset is the Brazilian E-Commerce Public Dataset by Olist, containing anonymized records of about 100,000 orders placed between 2016 and 2018 across multiple marketplaces in Brazil. It includes nine tables covering customers, orders, payments, items, products, categories, sellers, reviews, and geolocation data. Each table provides a different perspective, from customer demographics and purchase behavior to product attributes and delivery performance. While the dataset is robust, some fields contain missing or inconsistent values (e.g., product categories, delivery timestamps), so joins and aggregations must be handled carefully.

Three Business Insights

Insight 1 – Top Customers Drive Revenue

From the query identifying the top 10 customers by total spend, the highest spender contributed over 13,600 units, while the next few ranged between 7,200–6,900 units. This shows that a small group of customers contributes disproportionately to overall revenue. The business is highly dependent on these top spenders, which creates both opportunity (loyalty building) and risk (loss if they churn).

Insight 2 – Revenue Trends Show Growth and Seasonality

The monthly revenue trend query revealed steady growth from early 2017 through mid-2018, peaking at over 1.1M units in November 2017 and again in early 2018. However, dips in February and sharp month-to-month swings (as seen in the month-over-month change query) highlight strong seasonality and volatility. This means demand is predictable in peaks but unstable in off-season months, requiring careful planning.

Insight 3 – Product Categories Concentrated but Diverse

The product category revenue query showed that categories like bed, bath & table (1.71M units), health & beauty (1.65M units), and computers & accessories (1.58M units) dominate sales. Yet, the top category contributes only 8.54% of total revenue (from the category contribution query). This indicates that while a few categories are strong drivers, revenue is still spread across many smaller categories, balancing concentration with diversification.

Two Recommendations

Recommendation 1 – Strengthen Customer Retention

Since the repeat buyer query revealed that all 99,441 customers are one-time buyers, the business should implement loyalty programs, personalized offers, or subscription models to encourage repeat purchases. This will stabilize revenue, reduce reliance on constant new customer acquisition, and maximize the value of existing customers.

Recommendation 2 – Focus on High-Performing Categories and Products

Given that a handful of categories and top products dominate revenue, the business should prioritize marketing, inventory, and supplier relationships in these areas. At the same time, bundling or cross-selling strategies can be used to lift underperforming categories, ensuring growth is balanced and not overly reliant on a few segments.